

ST10

Space Technology

About 2-4 questions a paper, in every paper since 2013. The launch-vehicle-to-payload match and the ISRO centre-to-location match are near-certain. Add one current-affairs question on the newest mission.

India's space programme is the most reliably asked block in the Science and Technology section. RPSC does not test physics here - it tests names, dates, launch-vehicle-to-satellite matching and firsts. Learn it as a set of tables, not as a story.

Indian Space Programme

Basics

- Orbits - LEO, MEO, GEO, SSO, GTO
- Escape velocity 11.2 km/s
- Multi-stage rockets
- Cryogenic engines

Science missions

- Chandrayaan-1/2/3
- Mangalyaan
- AstroSat, Aditya-L1, XPoSat
- SpaDeX, NISAR

Policy and private

- Space Policy 2023
- IN-SPACe, NSIL, Antrix
- Skyroot, Agnikul, Pixxel
- Project NETRA, space debris

Institutions

- INCOSPAR 1962
- ISRO 15 Aug 1969
- Department of Space 1972
- VSSC, SDSC-SHAR, URSC, SAC, NRSC

Application satellites

- INSAT / GSAT
- IRS, Cartosat, Oceansat, RISAT
- NavIC / IRNSS
- Kalpana-1, EDUSAT

Launch vehicles

- SLV-3, ASLV
- PSLV and its variants
- GSLV Mk-II
- LVM3 / GSLV Mk-III
- SSLV, RLV-TD, HLVM3

Human spaceflight

- Gaganyaan on HLVM3
- TV-D1 crew escape test
- Vyommitra
- Bharatiya Antariksh Station 2035

The physics first - orbits, escape velocity, staging

Start with one idea. A satellite is not held up by anything. It is falling towards the Earth all the time, but it is moving sideways so fast that the Earth curves away beneath it at the same rate. That is an orbit - a permanent, controlled fall. Two speeds follow from this. Orbital velocity is the sideways speed needed to keep circling, about 7.9 km/s just above the atmosphere. Escape velocity is the speed at which you break free of Earth's gravity altogether - 11.2 km/s. Escape velocity does not depend on the mass of the object being launched; it depends on the mass and radius of the Earth.

- Orbital velocity near Earth's surface: about 7.9 km/s. Escape velocity: 11.2 km/s.
- Higher the orbit, slower the satellite and longer the period.
- At 35,786 km the period becomes 24 hours - the satellite then appears fixed over one point.
- Rockets are built in stages so that spent tanks and engines can be thrown away; less mass to accelerate means more speed from the fuel that remains. It is not about drag or heat.
- SLV-3 and ASLV were all-solid. PSLV alternates solid and liquid stages. Cryogenic stages sit on top, where efficiency matters most.

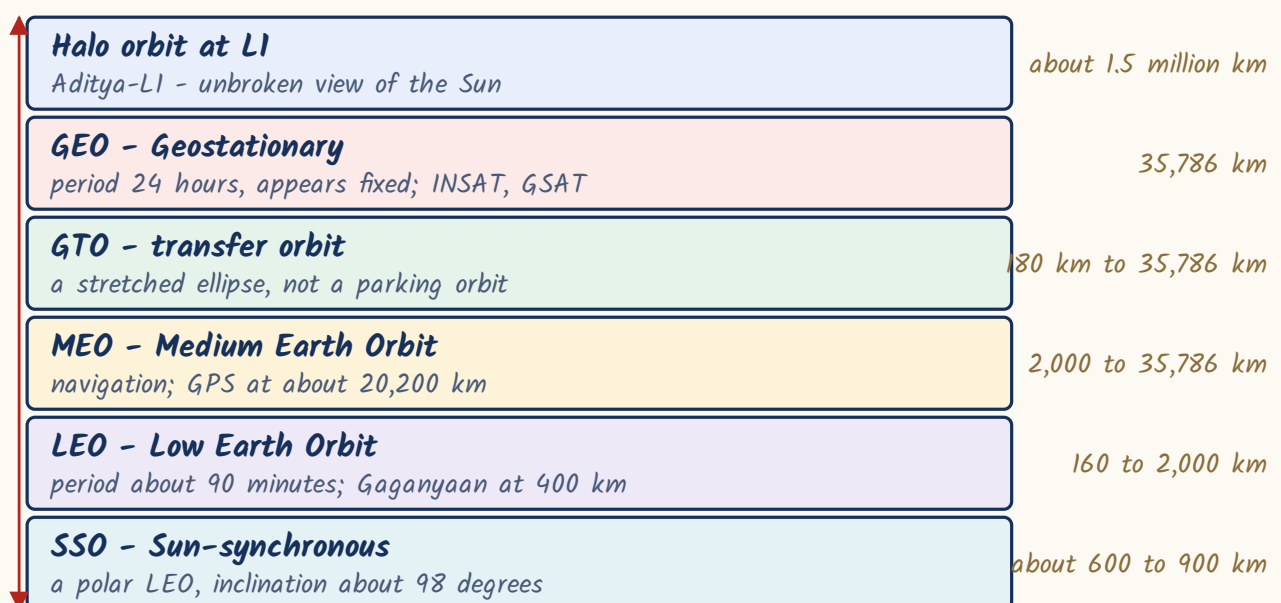
Orbit types - altitude, character, use

Orbit	Altitude / inclination	Character	Typical use
LEO - Low Earth Orbit	160 - 2,000 km	Period about 90 minutes	Earth observation, ISS, Gaganyaan at about 400 km
MEO - Medium Earth Orbit	Above LEO, up to 35,786 km	Period a few hours	Navigation constellations such as GPS at about 20,200 km
GEO - Geostationary	35,786 km, inclination about 0 degrees	Period about 24 hours, appears stationary	INSAT and GSAT communication and weather satellites

Orbit	Altitude / inclination	Character	Typical use
Polar / Sun-synchronous (SSO)	About 600-900 km, inclination about 98 degrees	Crosses a given latitude at the same local solar time every pass	IRS, Cartosat, RISAT, Resourcesat remote sensing
GTO - Geostationary Transfer Orbit	Perigee about 180 km, apogee about 35,786 km	A stretched transfer ellipse, not a parking orbit	Intermediate step before a satellite raises itself to GEO
Halo orbit at L1	About 1.5 million km, Sun-Earth L1	Balanced-gravity point with an uninterrupted view of the Sun	Aditya-L1

Understand why sun-synchronous matters, because RPSC has asked it as reasoning. The satellite passes over the same place at the same local solar time on every visit. So the Sun angle and the shadows are comparable from one pass to the next. That is exactly what you need to compare two images and detect change - a new building, a shrinking lake, a burnt forest. A geostationary satellite sits too far away to do this at high resolution.

The orbit ladder - by altitude



YAAD RAKHO

Ladder going upward: LEO - MEO - GEO. Sun-synchronous is a special polar LEO tilted at about 98 degrees. GTO is an ellipse used only to reach GEO. If an option calls GTO a 'circular orbit', it is wrong.

ISRO - the institution, its people and its centres

India's space programme began with a civilian, development-first idea: use space to solve problems on the ground - communication, weather, resource survey - not for prestige. Vikram Sarabhai set that direction. The institutional dates below are asked as straight one-liners, so lock them.

- INCOSPAR set up in 1962 under the Department of Atomic Energy.
- ISRO formed on 15 August 1969, superseding INCOSPAR.
- Department of Space and the Space Commission created in 1972.
- Vikram Sarabhai - father of the Indian space programme.
- Satish Dhawan chaired ISRO from 1972 to 1984; Sriharikota is named after him.
- V. Narayanan took charge as ISRO Chairman in January 2025.
- TERLS at Thumba, Thiruvananthapuram - chosen because the magnetic equator passes over it.
- First sounding rocket from Thumba: a Nike-Apache, launched 21 November 1963.

The centres and where they are

ISRO's principal centres

Centre	Location	What it does
VSSC - Vikram Sarabhai Space Centre	Thiruvananthapuram, Kerala	Design and development of launch vehicles

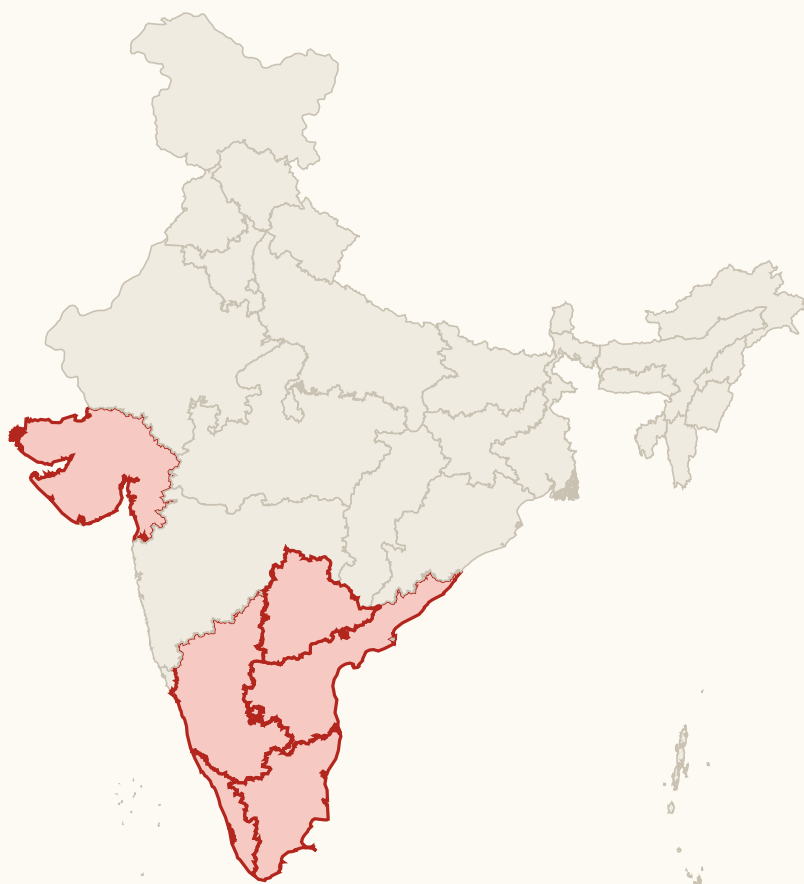
Centre	Location	What it does
LPSC - Liquid Propulsion Systems Centre	Valiamala (Thiruvananthapuram) and Bengaluru	Design of liquid propulsion stages
IPRC - ISRO Propulsion Complex	Mahendragiri, Tamil Nadu	Testing of liquid and cryogenic engines
SDSC-SHAR - Satish Dhawan Space Centre	Sriharikota, Andhra Pradesh	India's principal launch port
URSC - U R Rao Satellite Centre	Bengaluru, Karnataka	Building of satellites
SAC - Space Applications Centre	Ahmedabad, Gujarat	Payloads and space applications
NRSC - National Remote Sensing Centre	Hyderabad, Telangana	Reception and dissemination of remote sensing data
MCF - Master Control Facility	Hassan, Karnataka (also Bhopal)	Control of India's geostationary satellites
ISTRAC	Bengaluru, Karnataka	Tracking and low-Earth-orbit mission operations
Aeronautical Test Range	Chitradurga, Karnataka	RLV / Pushpak autonomous landing experiments

The supporting bodies - equally examinable

Body	Location	Role
PRL - Physical Research Laboratory	Ahmedabad, Gujarat	Space science research; runs the Udaipur Solar Observatory
NARL - National Atmospheric Research Laboratory	Gadanki, Andhra Pradesh	Atmospheric research
IIST - Indian Institute of Space Science and Technology	Valiamala, Thiruvananthapuram	India's first space university
HSFC - Human Space Flight Centre	Bengaluru, Karnataka	Runs the Gaganyaan human spaceflight programme

Body	Location	Role
IN-SPACe	Ahmedabad, Gujarat	Single-window authorisation and regulation of private space activity
NSIL - NewSpace India Limited	Bengaluru, Karnataka	Commercial arm of ISRO, created 2019

The states that hold ISRO's main centres



VSSC Kerala, IPRC Tamil Nadu, SDSC-SHAR Andhra, URSC and ISTRAC Karnataka, NRSC Telangana, SAC Gujarat

ASKED IN RAS

Asked repeatedly as a centre-to-location match - VSSC Thiruvananthapuram, SAC Ahmedabad, NRSC Hyderabad, URSC Bengaluru, IPRC Mahendragiri. And as a NOT-correctly-matched question: IIST is at Valiamala near Thiruvananthapuram, not Bengaluru.

Rajasthan link. The Udaipur Solar Observatory sits on an island in Fateh Sagar Lake. Water around it keeps the air steady, which reduces image shimmer - that is why an island was chosen. It was founded in 1976 by Arvind Bhatnagar, has been run by the Physical Research Laboratory, Ahmedabad, since 1981, and its 50 cm Multi Application Solar Telescope (MAST) became operational in 2015.

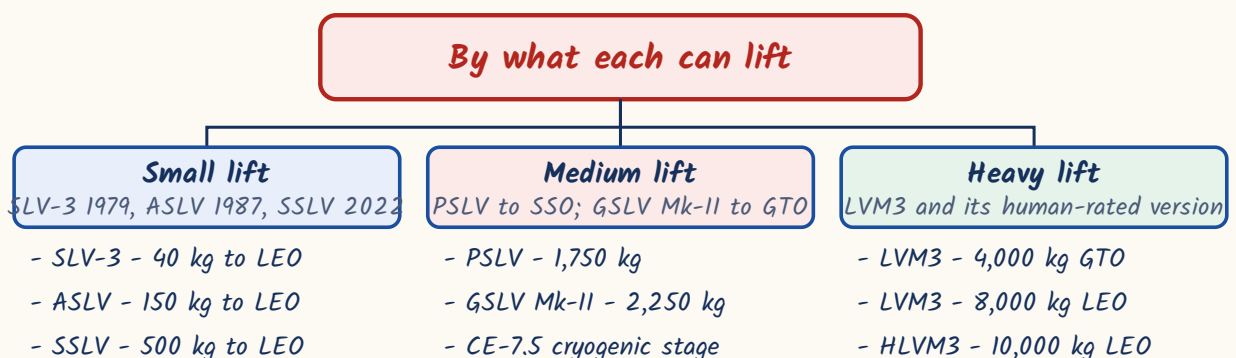
ASKED IN RAS

Asked in RAS Pre 2018 and again in RAS Pre 2021, in near-identical wording - the Udaipur Solar Observatory is on an island in Fateh Sagar Lake. A repeat this exact is a gift; do not lose it.

The launch vehicles - the table RPSC keeps returning to

A launch vehicle is judged by two numbers: how much it can lift, and to which orbit. A rocket that can put 1,750 kg into a low polar orbit will manage far less into a geostationary transfer orbit, because height costs energy. Read the table with that in mind, and memorise the mission column - that is where the marks are.

Indian launch vehicles by lift class



Indian launch vehicles - payload capacity and the missions each carried

Vehicle (first flight)	Configuration	Payload capacity	Key missions
SLV-3 (1979)	Four stages, all solid	40 kg to LEO	Rohini RS-1, 18 July 1980; project director A.P.J. Abdul Kalam
ASLV (1987)	Five stages, all solid	150 kg to LEO	SROSS series of experimental satellites
PSLV (1993)	Four stages, solid and liquid alternating	About 1,750 kg to a 600 km sun-synchronous orbit	Chandrayaan-1, Mangalyaan, AstroSat, Aditya-L1, XPoSat, SpaDeX, Cartosat, RISAT
GSLV Mk-II (2001)	Three stages, indigenous CE-7.5 cryogenic upper stage	About 2,250 kg to GTO	EDUSAT (GSLV-F01), NVS-02 navigation satellite, NISAR (GSLV-F16)
LVM3 / GSLV Mk-III (2014, experimental)	Three stages, CE-20 cryogenic upper stage	4,000 kg to GTO, 8,000 kg to LEO	Chandrayaan-2, Chandrayaan-3, OneWeb constellation, CMS-03, BlueBird Block-2
SSLV (2022)	Three solid stages plus a liquid velocity trimming module	About 500 kg to a 500 km LEO	EOS-08 on SSLV-D3, August 2024
RLV-TD / Pushpak	Winged reusable technology demonstrator	Experimental - no operational payload	HEX-01 in May 2016; LEX autonomous landings at Chitradurga
HLVM3	Human-rated LVM3 with a crew escape system	About 10,000 kg to LEO	Gaganyaan uncrewed and crewed flights

PSLV is called ISRO's workhorse because it flies almost anywhere - polar orbits, transfer orbits, even out to the Moon and Mars. Its flexibility comes from strap-on boosters that can be added or removed, and the variant name simply tells you how many. Learn this; it converts a hard question into an easy one.

PSLV variants - the letters count the strap-ons

Variant	Strap-on boosters	Note
PSLV-G (generic)	6	The original configuration, now retired
PSLV-CA	0 - 'core alone'	Lightest version, used for smaller payloads
PSLV-DL	2	Introduced with PSLV-C44
PSLV-QL	4	Introduced with PSLV-C45
PSLV-XL	6 extended-length boosters	Heaviest version - carried Chandrayaan-1 and Mangalyaan

- PSLV-D1 in 1993 failed; the first fully successful flight was PSLV-D2 in 1994.
- PSLV-C37, 15 February 2017 - world record of 104 satellites in a single flight, primary payload Cartosat-2 series.
- That record was surpassed by SpaceX's Transporter-1 with 143 satellites in January 2021.
- PSLV-C55, April 2023 - launched Singapore's TeLEOS-2 as a dedicated commercial mission for NSIL.
- SSLV-D1, August 2022, failed to reach a stable orbit; SSLV-D2 (February 2023) and SSLV-D3 (August 2024) succeeded.
- SSLV technology has been transferred to HAL; a second launch port for SSLV is coming up at Kulasekarapattinam, Thoothukudi district, Tamil Nadu.
- GSLV-F15 carrying NVS-02, on 29 January 2025, was ISRO's 100th launch from Sriharikota.

PSLV versus LVM3 - the swap RPSC exploits

PSLV	LVM3 / GSLV Mk-III
Four stages, solid and liquid alternating	Three stages, two solid boosters plus a cryogenic upper stage
About 1,750 kg to sun-synchronous orbit	4,000 kg to GTO and 8,000 kg to LEO
Carried Chandrayaan-1 (2008) and Mangalyaan (2013)	Carried Chandrayaan-2 (2019) and Chandrayaan-3 (2023)
Aditya-L1, XPoSat, SpaDeX, AstroSat	OneWeb constellation, CMS-03, BlueBird Block-2
ISRO's workhorse	India's heaviest operational rocket; being human-rated as HLVM3

ASKED IN RAS

Two straight repeats. RAS Pre 2018 - PSLV-C37 launched 104 satellites in February 2017. RAS Pre 2023 - TeLEOS-2 was launched by PSLV-C55. RAS Pre 2023 also gave a mission-to-launch-vehicle match: Chandrayaan-1 to PSLV-C11, Mangalyaan to PSLV-C25, Chandrayaan-3 to LVM3-M4, Aditya-L1 to PSLV-C57. Learn the C-numbers - RPSC puts them in the options.

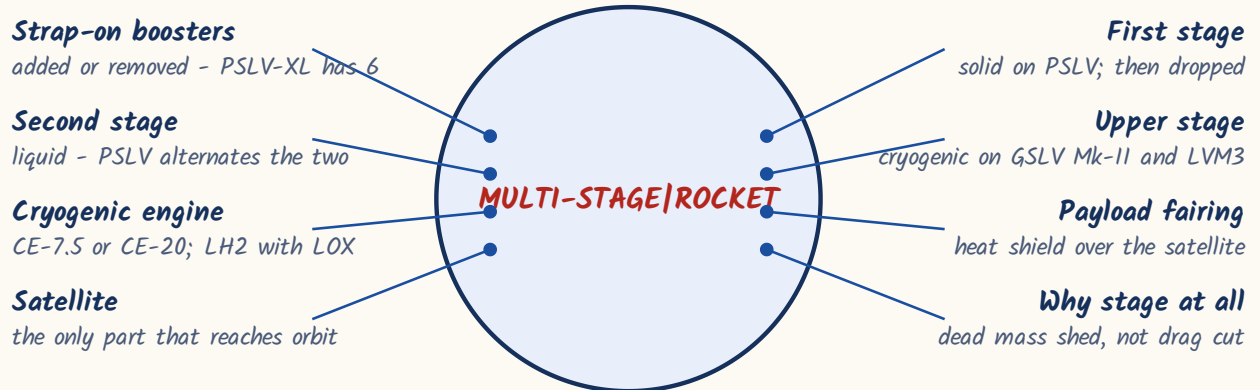
TRAP

Chandrayaan-1 and Mangalyaan flew on PSLV; Chandrayaan-2 and Chandrayaan-3 flew on GSLV Mk-III / LVM3. Every year somebody swaps these. Second trap in the same family: SLV-3 was all-solid and never had a cryogenic stage - India's first indigenous cryogenic flight was GSLV-D5 in January 2014.

Cryogenic engines and reusability

A cryogenic engine burns liquid hydrogen with liquid oxygen. Both must be held at extremely low temperatures - hydrogen below about minus 253 degrees Celsius. That is the hard part: pumps, seals and tanks that survive those temperatures. The reward is the highest efficiency of any chemical propellant, so cryogenics go in the upper stages. India was denied the technology in the 1990s and built it in-house, which is why the exam keeps asking about it.

How a multi-stage launch vehicle is built



- CE-7.5 - indigenous cryogenic engine on GSLV Mk-II.
- CE-20 - the larger cryogenic engine on LVM3.
- GSLV-DS, January 2014 - India's first successful flight with an indigenous cryogenic stage.
- SCE-200 semi-cryogenic engine, burning kerosene with liquid oxygen, is under development.
- RLV-TD is ISRO's winged reusable demonstrator; HEX-01 flew in May 2016 and the Pushpak (LEX) landing experiments were done at the Aeronautical Test Range, Chitradurga, Karnataka.

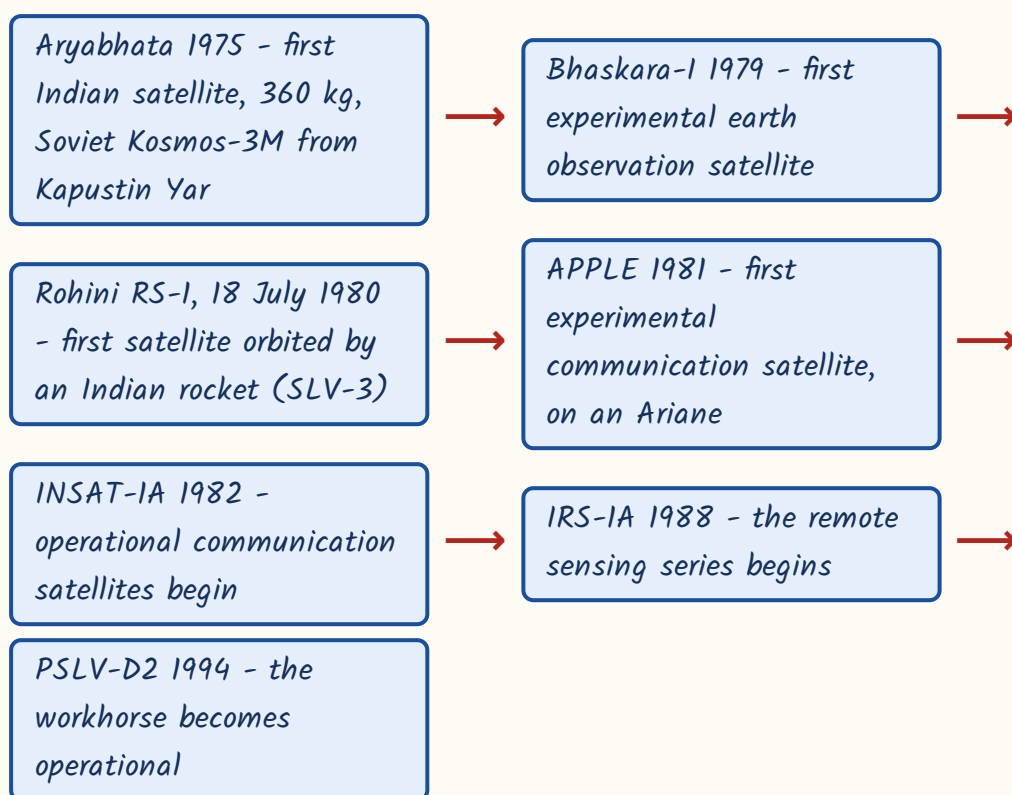
YAAD RAKHO

Cryogenic means liquid hydrogen plus liquid oxygen. Semi-cryogenic means kerosene plus liquid oxygen. Any option that speaks of a 'solid cryogenic engine' is wrong by definition.

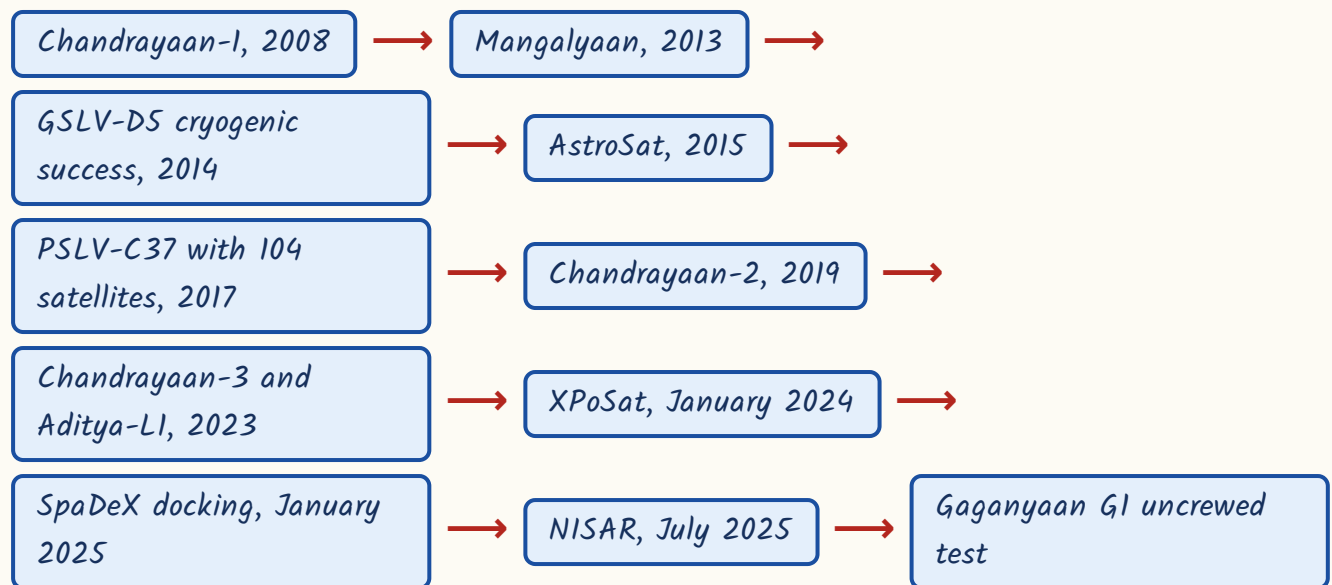
The milestones - lock the chronology

RPSC asks chronological ordering in this chapter. The safe method is to fix a handful of anchor years and place everything else against them: 1975 Aryabhata, 1980 Rohini, 2008 Chandrayaan-1, 2013 Mangalyaan, 2023 Chandrayaan-3.

The foundation years



The modern milestones



- *Aryabhata - India's first satellite, 360 kg, launched 19 April 1975 from Kapustin Yar in the USSR.*
- *Rohini RS-1 - first Indian satellite placed in orbit by an Indian launch vehicle.*
- *APPLE, 1981 - first experimental communication satellite, launched by an Ariane rocket.*
- *Kalpana-1, 2002, originally METSAT - India's first dedicated meteorological satellite, renamed after Kalpana Chawla.*
- *EDUSAT, 2004 on GSLV-F01 - first Indian satellite built exclusively for education.*
- *Remember the trio A-R-A: Aryabhata on a Soviet rocket, Rohini on an Indian rocket, APPLE on a European rocket.*

Deep space and science missions - the highest-yield block

These missions carry the 'firsts' examiners love. For each one hold four things: launch date, launch vehicle, what it achieved, and one distinguishing detail.

Chandrayaan-1 (2008) was an orbiter that also dropped a Moon Impact Probe. Its most quoted result did not come from an Indian instrument - NASA's Moon Mineralogy Mapper, riding on the Indian orbiter, confirmed water molecules on the lunar surface.

- Chandrayaan-1 - October 2008 on PSLV-C11; Moon Impact Probe; confirmed water molecules on the Moon.
- Chandrayaan-2 - 22 July 2019 on GSLV Mk III-M1; the Vikram lander crashed on 7 September 2019; the orbiter still works.
- The Chandrayaan-2 crash site was named Tiranga Point.
- Chandrayaan-3 - 14 July 2023 on LVM3-M4; Vikram soft-landed on 23 August 2023 near the lunar south pole.
- India became the 4th country to soft-land on the Moon and the 1st to land near the lunar south pole.
- The Chandrayaan-3 landing site is Shiv Shakti Point; 23 August is now National Space Day.
- Chandrayaan-3 cost about Rs 615 crore.

ASKED IN RAS

Asked in RAS Pre 2023 - Vikram soft-landed on 23 August 2023. A second standard one-liner: Chandrayaan-1 is chiefly credited with the discovery of water molecules on the lunar surface. Do not attribute that discovery to Chandrayaan-2.

Science missions at a glance

Mission	Launched on	Domain and achievement
Mangalyaan (Mars Orbiter Mission)	5 November 2013, PSLV-C25	Mars orbit insertion 24 September 2014; India the first country to reach Mars orbit on its maiden attempt; cost about Rs 450 crore

Mission	Launched on	Domain and achievement
AstroSat	2015, PSLV-C30	India's first multi-wavelength space observatory
Aditya-L1	2 September 2023, PSLV-C57	Solar mission; halo orbit around Sun-Earth L1 from 6 January 2024; main payload VELC built by the Indian Institute of Astrophysics
XPoSat	1 January 2024, PSLV-C58	World's second X-ray polarimetry mission after NASA's IXPE; payloads POLIX and XSPECT
SpaDeX	30 December 2024, PSLV-C60	SDX01 Chaser and SDX02 Target docked on 16 January 2025 - India the 4th country to dock in space after the USA, Russia and China
NISAR	30 July 2025, GSLV-F16	ISRO-NASA dual-band synthetic aperture radar - NASA L-band plus ISRO S-band, in a sun-synchronous orbit

ASKED IN RAS

Asked in RAS Pre 2016 - Mangalyaan entered Mars orbit on 24 September 2014. Carry both dates: launch 5 November 2013, insertion 24 September 2014. Match the mission to what it looks at - AstroSat many wavelengths, Aditya-L1 the Sun, XPoSat X-ray polarisation, NISAR radar imaging of the Earth, SpaDeX docking.

Application satellites, remote sensing and GIS

This is the part of the programme that pays for itself. Communication satellites carry television, telephony and disaster warnings. Earth observation satellites survey crops, water, forests and coastlines. Navigation satellites give position. RPSC asks these as series-to-purpose matches, so learn the family names and one line each.

The satellite families and what each does

Series	Orbit	Principal application
<i>INSAT and GSAT</i>	<i>Geostationary</i>	<i>Communication, broadcasting and meteorology</i>
<i>IRS - Indian Remote Sensing</i>	<i>Sun-synchronous polar</i>	<i>Earth observation; the series began with IRS-1A in 1988</i>
<i>Cartosat</i>	<i>Sun-synchronous polar</i>	<i>High-resolution cartographic mapping; Cartosat-3 gives about 0.25 m panchromatic resolution</i>
<i>Resourcesat</i>	<i>Sun-synchronous polar</i>	<i>Land use, agriculture and natural resource survey</i>
<i>Oceansat</i>	<i>Sun-synchronous polar</i>	<i>Ocean colour and ocean surface wind studies</i>
<i>RISAT</i>	<i>Sun-synchronous polar</i>	<i>All-weather synthetic aperture radar imaging - sees through cloud and at night</i>
<i>EOS series</i>	<i>Mostly sun-synchronous</i>	<i>Renamed earth observation satellites, for example EOS-08 on SSLV-D3</i>
<i>NavIC / IRNSS</i>	<i>Geosynchronous and geostationary</i>	<i>Regional navigation - designed as a constellation of 7 satellites</i>
<i>Kalpana-1, 2002</i>	<i>Geostationary</i>	<i>First dedicated Indian meteorological satellite, originally METSAT</i>
<i>EDUSAT, 2004</i>	<i>Geostationary</i>	<i>First Indian satellite exclusively for education</i>

- NavIC, earlier IRNSS, covers India and about 1,500 km beyond India's borders - regional, not global.
- GAGAN is a satellite-based augmentation system built jointly by ISRO and the Airports Authority of India to sharpen GPS accuracy for civil aviation. NavIC and GAGAN are two different systems.
- Remote sensing means recording the electromagnetic radiation an object reflects or emits, without touching it.
- Passive sensors read reflected sunlight or emitted heat; active sensors such as radar send their own signal and read the echo - which is why RISAT works through cloud and at night.
- Spatial resolution is the smallest object distinguishable; spectral resolution is the number and width of bands; temporal resolution is the revisit interval.
- GIS - a Geographic Information System - stores map layers and analyses the relations between them; it is the software that turns imagery into a decision.
- Indian uses: crop forecasting, groundwater prospecting, forest cover mapping, urban planning, disaster damage assessment, and the Bhuvan geoportal.

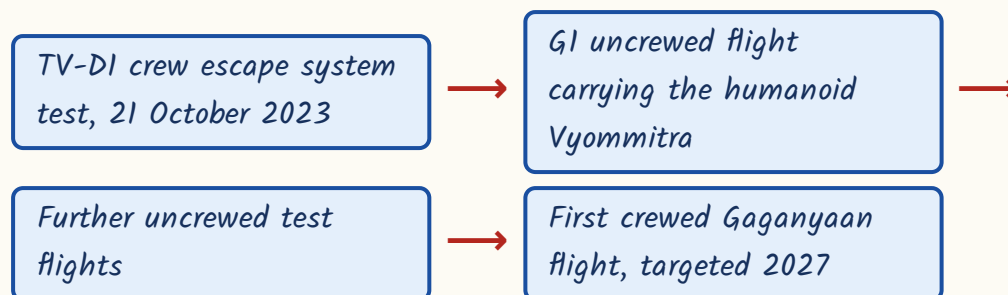
TRAP

Kalpana-1 is meteorological, not navigational - navigation is NavIC / IRNSS. RPSC has used exactly this NOT-correctly-matched pair. Second confusion in the same set: Cartosat is mapping, RISAT is radar, Oceansat is ocean colour.

Gaganyaan and human spaceflight

Gaganyaan is India's crewed spaceflight programme. Three crew, an orbit of about 400 km, a mission of up to seven days, and a safe splashdown. The launcher is HLV M3 - the LVM3 rebuilt to human-rating standards and fitted with a crew escape system that can pull the crew module clear of a failing rocket. Approved outlay about Rs 20,193 crore; the programme is run from the Human Space Flight Centre, Bengaluru.

The Gaganyaan sequence



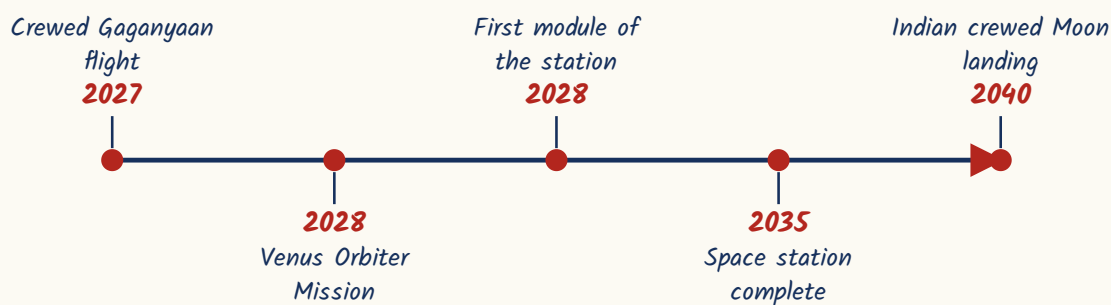
CURRENT LINK

Current link - as of September 2026 the uncrewed Gaganyaan G1 mission had not yet flown; ISRO has been targeting the second half of 2026. The crew module completed an air-drop test from an Indian Air Force Chinook in April 2026, and Vyommitra was being integrated into the crew module. Expect a question on the launcher name (HLVM3) and on Vyommitra.

Two Indians have been to space. Rakesh Sharma flew aboard Soyuz T-11 in 1984. Group Captain Shubhanshu Shukla flew the Axiom-4 mission to the International Space Station in June 2025 - the second Indian in space and the first Indian to visit the ISS.

- The Union Cabinet cleared a long-term package on 18 September 2024 - learn the three names, costs and target years together.
- Chandrayaan-4 - lunar sample return mission, about Rs 2,104 crore.
- Venus Orbiter Mission (Shukrayaan) - about Rs 1,236 crore, targeted for 2028.
- Bharatiya Antariksh Station - first module by 2028, the station complete by 2035.
- Indian crewed landing on the Moon - target 2040.

The road ahead - the declared targets



Space policy, private players, debris and the world outside

Until 2020, space in India meant ISRO and nothing else. The Indian Space Policy 2023 changed the architecture: ISRO concentrates on research and advanced missions, while private companies build and fly. Three names carry the new structure, and RPSC has already begun asking about them.

Who does what after the 2023 policy

Body	Set up	Role
IN-SPACe	2020, headquartered at Ahmedabad	Single-window autonomous agency that authorises and regulates non-government space activity

Body	Set up	Role
NSIL - NewSpace India Limited	2019, Bengaluru	Commercial PSU arm - owns and sells launches and satellite services
Antrix Corporation	1992	The earlier marketing arm of ISRO - not to be confused with NSIL

The private players you should be able to name

Company	Product	Distinction
Skyroot Aerospace	Vikram-S sounding rocket	Mission Prarambh, 18 November 2022 - India's first privately built rocket
Agnikul Cosmos	Agnibaan SOrTeD	May 2024 - the world's first single-piece 3D-printed semi-cryogenic engine
Pixxel	Hyperspectral imaging satellites	Commercial earth observation
Dhruva Space	Satellite orbital deployers	Small satellite platforms and deployment hardware

- Every dead satellite and spent rocket stage stays in orbit; at orbital speed even a fleck of paint can hole a spacecraft, and one collision creates thousands more fragments.
- Project NETRA is ISRO's space object tracking initiative.
- IS40M, at Bengaluru, is the ISRO System for Safe and Sustainable Space Operations Management.
- ISRO's stated goal is debris-free space missions by 2030.

International programmes - the minimum you must know

Agency / programme	Country	Key fact
NASA - Artemis	United States	Artemis-1 flew uncrewed around the Moon in November-December 2022
James Webb Space Telescope	NASA with ESA and CSA	Launched December 2021 on an Ariane-5; operates from the Sun-Earth L2 point

Agency / programme	Country	Key fact
International Space Station	USA, Russia, ESA, Japan, Canada	In low Earth orbit; Shubhanshu Shukla visited on Axiom-4 in 2025
CNSA - Chang'e programme	China	Chang'e-6 returned the first samples from the far side of the Moon in June 2024
Roscosmos	Russia	Launched Aryabhata in 1975 and Rakesh Sharma on Soyuz T-11 in 1984
JAXA	Japan	The Japanese space agency
CNES	France	The French space agency - Canada's is the Canadian Space Agency, CSA
SpaceX - Falcon 9, Starship	United States, private	Transporter-1 set the 143-satellite record in January 2021

TRAP

CNES is French, not Canadian. Antrix is 1992, NSIL is 2019. IN-SPACe regulates and authorises; NSIL sells. These three swaps carry almost every NOT-correctly-matched question in this section.

CURRENT LINK

Current link, and read the wording carefully. CMS-03, launched by LVM3-M5 on 2 November 2025 and weighing about 4,410 kg, was the heaviest satellite launched from Indian soil AT THE TIME. It was surpassed on 24 December 2025, when LVM3-M6 placed the roughly 6,100 kg BlueBird Block-2 satellite in orbit for AST SpaceMobile. So an unqualified option saying CMS-03 is 'the heaviest ever launched from India' is wrong - as of September 2026 the holder is BlueBird Block-2.

REVISION RECAP

1. Escape velocity 11.2 km/s, orbital velocity about 7.9 km/s, GEO at 35,786 km with a 24-hour period.

2. Sun-synchronous polar orbit, inclination about 98 degrees - used for remote sensing because the local solar time stays constant.
3. Staging works by shedding dead mass, not by reducing drag.
4. INCOSPAR 1962, ISRO 15 August 1969, Department of Space 1972; Sarabhai the father, Satish Dhawan chairman 1972-84, V. Narayanan chairman from January 2025.
5. Thumba TERLS - Nike-Apache sounding rocket, 21 November 1963, site chosen for the magnetic equator.
6. VSSC Thiruvananthapuram, SAC Ahmedabad, NRSC Hyderabad, URSC Bengaluru, IPRC Mahendragiri, MCF Hassan, IIST Valiamala.
7. Udaipur Solar Observatory - island in Fateh Sagar Lake, founded 1976 by Arvind Bhatnagar, run by PRL, 50 cm MAST since 2015.
8. Payloads: SLV-3 40 kg, ASLV 150 kg, PSLV about 1,750 kg to SSO, GSLV Mk-II about 2,250 kg to GTO, LVM3 4,000 kg to GTO, SSLV about 500 kg to LEO, HLV3 about 10,000 kg to LEO.
9. PSLV variant names count the strap-ons - CA zero, DL two, QL four, XL six.
10. PSLV-C37 put up 104 satellites on 15 February 2017; beaten by SpaceX Transporter-1 with 143 in January 2021.
11. Launch pairs: Chandrayaan-1 PSLV-C11, Mangalyaan PSLV-C25, AstroSat PSLV-C30, Aditya-L1 PSLV-C57, XPoSat PSLV-C58, SpaDeX PSLV-C60, Chandrayaan-3 LVM3-M4, NISAR GSLV-F16.
12. CE-7.5 on GSLV Mk-II, CE-20 on LVM3; the first indigenous cryogenic success was GSLV-DS in January 2014.
13. Firsts: Aryabhata 1975 Soviet rocket, Rohini RS-1 1980 SLV-3, APPLE 1981 Ariane, Kalpana-1 2002 weather, EDUSAT 2004 education.
14. Chandrayaan-3 landed on 23 August 2023 at Shiv Shakti Point - 4th nation to soft-land, 1st near the lunar south pole; 23 August is National Space Day.
15. Mangalyaan reached Mars orbit on 24 September 2014 at the first attempt, at about Rs 450 crore.
16. SpaDeX docked on 16 January 2025, making India the 4th docking nation; NISAR is the ISRO-NASA dual-band radar mission of July 2025.
17. NavIC is a 7-satellite regional system covering India plus 1,500 km; GAGAN is the ISRO-AAI augmentation system for aviation.
18. Gaganyaan - Rs 20,193 crore, HLV3 launcher, 3 crew, about 400 km, up to 7 days, Vyommitra on the uncrewed flight, crewed flight targeted 2027.
19. Cabinet of 18 September 2024 - Chandrayaan-4 sample return, Venus Orbiter for 2028, Bharatiya Antariksh Station complete by 2035, crewed Moon landing 2040.
20. IN-SPACe authorises from Ahmedabad, NSIL sells since 2019, Antrix was the 1992 marketing arm; Vikram-S was India's first private rocket, November 2022.